Abhishek Santhakumar

ML Engineer
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Education

State University of New York, Albany **GPA - 3.25**
Master of Science in Computer Science Aug 2024 - May 2026
Scholarship - Dean's Merit Scholarship

Rajalakshmi Institute of Technology **GPA - 3.24**
Bachelor's in computer science and engineering Aug 2019 - May 2023

Professional Experience

Research Assistant | Part-time
University At Albany **Feb 2026 - Present**

- Designed and implemented evaluation frameworks in a Unix environment using Python scripts to analyze academic datasets, which improved consistency of research outcomes across projects
- Collected, processed, and analyzed research data with Bash and Python tools, ensuring accuracy and compliance with academic standards, which enabled timely completion of the study
- Assisted in experimental studies and lab sessions by guiding undergraduates through C and PHP programming exercises, helping them grasp core concepts and improve problem-solving skills
- Provided one-on-one academic support in Java and object-oriented programming, using code reviews and debugging sessions to clarify concepts, which increased student confidence in OOP

Associate Engineer | Full-time
L&T Technology Services **July 2023 - June 2024**

- Developed and maintained automated testing frameworks in Python, improving test efficiency and reducing manual testing efforts.
- Led quality assurance initiatives that reduced production defects by 40%, leveraging enhanced test coverage, performance monitoring, and debugging strategies.
- Diagnosed and resolved critical system issues with log analysis and debugging tools, maintaining 99.9% system uptime and meeting business KPI targets
- Collaborated with cross-functional teams in an Agile environment to plan sprints, groom backlogs, and deliver 12 feature releases on schedule, improving delivery predictability
- Supported migration of cloud-based applications and PostgreSQL databases to AWS RDS and Redshift, validating data integrity and completing the transition with zero data loss and minimal downtime
- Monitored system performance using PostgreSQL's pg_stat_statements and custom C# scripts, tracked key metrics such as query latency and CPU usage, identified bottlenecks, and applied query-optimization and data-modeling changes that improved overall system response time

Android Application developer | Intern
Thinq24 Innovations Pvt Ltd **Dec 2022 - Jan 2023**

- Designed and developed Android applications with a focus on performance, usability, and responsive UI/UX.
- Built reusable UI components and implemented scalable features aligned with client requirements.
- Integrated RESTful APIs to enable efficient communication between frontend and backend systems.
- Collaborated with backend developers to ensure seamless data flow and application performance optimization.
- Tested and debugged applications to enhance stability and deliver a high-quality user experience.

Skills

Programming & Development: Python, Java, C, C++, C#, JavaScript (JS), TypeScript, Node.js, React, HTML, CSS, SQL, PHP, Kotlin, Ruby.
Visual Basic Core Programming Languages: Object-oriented and procedural programming, data structures, and algorithms.
Frontend & UI: React.js, Vue.js.
Databases: MySQL, Oracle, PostgreSQL, MongoDB, schema design, data modeling, query optimization.
Tools & Platforms: Git, Android Studio, Linux, Unix environments, LaTeX, NetworkX.
Cloud & Infrastructure: AWS (S3, RDS, Redshift, SQS, SNS concepts), Microsoft Azure (fundamentals).

Academic Projects

Neural Networks for Classification

- Developed and optimized multi-layer neural networks using Python and TensorFlow/Keras for classification tasks across structured and unstructured datasets.
- Improved model performance through hyperparameter tuning (learning rate, batch size), regularization (L1/L2), and early stopping, reducing overfitting and increasing generalization.
- Evaluated models using accuracy, precision, recall, F1-score, and confusion matrices to ensure robustness.
- Implemented data preprocessing pipelines, including normalization, feature scaling, and handling missing values.

Lung Cancer Detection using CNN

- Built a Convolutional Neural Network (CNN) model for early detection of lung cancer from medical imaging datasets.
- Applied data augmentation techniques (rotation, flipping, scaling) to improve model generalization on limited datasets.
- Designed deep learning architectures with convolutional, pooling, and fully connected layers, optimizing feature extraction.
- Improved performance using dropout, batch normalization, and learning rate scheduling.

Graph Neural Network-based Social Network Analysis

- Applied graph-based ML techniques using Python and NetworkX to analyze large-scale social network data.
- Implemented link prediction algorithms (Jaccard Index, Adamic/Adar, Degree Product) to infer missing edges in sparse graphs.
- Performed community detection (label propagation) and analyzed homophily patterns to understand user clustering behavior.